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**Nepali Audio Deepfake Detection Using Self-Supervised Speech
Representations**

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ABSTRACT

The rapid advancement of text-to-speech, voice conversion, and voice cloning technologies has made synthetic speech increasingly natural and difficult to distinguish from genuine human speech. While audio deepfake detection has been widely studied for high-resource languages such as English and Chinese, comparatively little work has addressed low-resource languages such as Nepali. This proposal presents a research-based experimental study for Nepali audio deepfake detection using multilingual self-supervised speech representations and graph-based spoofing analysis. The proposed system will construct a speaker-disjoint Nepali real/fake speech dataset from public or consented bona fide speech and synthetic speech generated through TTS and voice-cloning tools. An English-trained baseline detector will first be evaluated on Nepali audio to quantify cross-lingual degradation. The detector will then be adapted using XLS-R front-end, parameter-efficient fine-tuning, and an AASIST spectro-temporal graph-attention back-end. Performance will be evaluated using Equal Error Rate, accuracy, precision, recall, F1-score, and confusion matrix analysis. The expected outcome is a Nepali-focused audio deepfake detection framework and dataset preparation pipeline that addresses the current language gap in audio forensics research.

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ABBREVIATIONS

ADD	Audio Deepfake Detection
AASIST	Audio Anti-Spoofing using Integrated Spectro-Temporal Graph Attention Networks
ASV	Automatic Speaker Verification
ASVspoof	Automatic Speaker Verification Spoofing and Countermeasures Challenge
CNN	Convolutional Neural Network
EER	Equal Error Rate
FC	Fully Connected
LFCC	Linear Frequency Cepstral Coefficient
LoRA	Low-Rank Adaptation
MFCC	Mel-Frequency Cepstral Coefficient
PEFT	Parameter-Efficient Fine-Tuning
RNN	Recurrent Neural Network
SSL	Self-Supervised Learning
TTS	Text-to-Speech
VC	Voice Conversion
XLS-R	Cross-Lingual Speech Representation

CHAPTER 1

INTRODUCTION

Advancement in speech and audio generation technologies have increased the quality and naturalness of the synthesized audio. Numerous technologies have emerged for this case. Emergence of Text-to-Speech (TTS), Voice Conversion (VC) and voice cloning have enabled numerous beneficial applications in accessibility, virtual assistants and human-computer interaction along with misuse to create deceptive audio content which are referred to as deepfake audio [16, 17]. Deepfake audio are convincing enough to mimic human voices, posing significant threat to the integrity of digital content and worsening the challenges of misinformation. This proliferation of highly realistic synthesized speech which are referred as deepfakes has necessitated robust countermeasure, the ability to detect deepfake audio.

Audio deepfake detection (ADD) is a task that aims to distinguish genuine utterances from fake ones via machine learning techniques [9, 10]. Audio deepfake detection uses machine learning and acoustic analysis to identify synthesized audios. While substantial research has been conducted on audio deepfake detection for high-resource languages such as English and Chinese, comparatively little work has focused on low-resource languages such as Nepali. Existing state-of-the-art models are primarily trained and evaluated using datasets collected from high-resource languages, making their effectiveness in the Nepali language domain largely unexplored. Furthermore, publicly available Nepali deepfake detection datasets remain extremely limited, creating additional challenges for developing robust and language-specific detection systems.

Since the robust architectures has been known to have matured in cloning Nepali Voice, the defense mechanism is lacking, thus, this project proposes a robust, localized audio deepfake detection framework for Nepali speech using multilingual self-supervised learning (SSL) representations. While state-of-the-art detectors achieve near-zero error rates on dominant Western benchmarks, literature confirms a catastrophic performance collapse when these English-trained models encounter out-of-distribution, low-resource regional languages [10, 11]. To resolve the issue, a resilient audio deepfake detection system for Nepali speech, pairing genuine human speech harvested from public media with synthetic voice clones generated via modern multilingual Text-to-Speech (TTS)

and Voice Conversion (VC) tools is proposed. Effectiveness of Low-Rank Adaptation (LoRA), a parameter-efficient fine-tuning technique, applied to a pre-trained XLS-R (Wav2Vec2) front-end coupled with Audio Anti-Spoofing Using Integrated Spectro-Temporal Graph Attention Networks (AASIST) back-end is investigated.

CHAPTER 2

PROBLEM STATEMENT

The rapid advancement of speech synthesis including Text-to-Speech (TTS), Voice Conversion (VC), and Zero-shot voice cloning has enabled generation of convincing deepfake audio, posing serious threats to digital audio authenticity and enabling identity fraud and misinformation. Significant amount of work has been done on the audio deepfake detection for high resource language with comparatively low work for Nepali speech. Compounding to that, no standardize dataset currently exists for Nepali Speech.

As an effort toward this, a fine-tuned model from the 128-language pre-trained XLS-R front-end coupled with AASIST back-end for Nepali Speech with the data to be harvested from the public media and synthetic voice cloning can be a solution.

CHAPTER 3

OBJECTIVES

3.1 Main Objective

The main objective of this project is to fine-tune audio deepfake detection system for Nepali Speech leveraging multilingual self-supervised deep learning-based classification techniques.

3.2 Specific Objectives

The specific objectives of this project are:

- (i) To harvest a labeled Nepali speech dataset comprising of genuine speech collected from verified public media sources and synthetic speech generated through TTS and voice cloning system.
- (ii) To preprocess both existing and newly collected audio data to ensure an acoustically balanced training corpus.
- (iii) To fine tune XLS-R self-supervised front end coupled and AASIST graph attention backend with LoRA-based Parameter Efficient Fine Tuning (PEFT).
- (iv) To evaluate the performance and robustness of proposed system using appropriate evaluation metrics such as Equal Errors Rate (EER), classification accuracy, etc.

CHAPTER 4

LITERATURE REVIEW

4.1 Review of Related Theory

4.1.1 Audio Spoofing Paradigms: Text-to-Speech and Voice Conversion

The automatic speaker verification spoofing countermeasure literature, formalised through the ASVspoof challenge series, divides attacks into logical access, physical access or replay, deepfake, and adversarial categories [1–4]. This project is concerned with the logical-access and deepfake setting, where a synthesised or converted waveform is directly submitted to a verification or listening system. This setting has become the most urgent for low-resource languages because modern public text-to-speech and voice-cloning systems make the attack feasible without specialised recording equipment.

Two attack families dominate this setting. In text-to-speech (TTS), a transcript is converted into a waveform that imitates the target speaker. In voice conversion (VC), a source utterance is transformed so that it preserves the linguistic content of the speaker but changes the speaker identity. The ASVspoof 2019 Logical Access database deliberately combined multiple TTS and VC systems so that detectors would be evaluated against unseen synthesizers rather than against only a memorised attack pattern [3, 18]. This generalisation requirement is central to the present work, where an English-trained reference detector is expected to face Nepali speech and Nepali synthesis artefacts.

The detection problem can be expressed as a binary hypothesis test on an input utterance x :

$$H_0 : x \in B \text{ (bona fide)}, \quad H_1 : x \in S \text{ (spoofed)} \quad (4.1)$$

Here, B denotes the set of genuine utterances and S denotes the set of spoofed utterances. A countermeasure produces a score $s(x)$ and classifies the utterance by thresholding:

$$\hat{y} = \mathbb{1}[s(x) \geq \tau] \quad (4.2)$$

where τ is the decision threshold and $\hat{y} = 1$ denotes a spoof decision. Equal Error Rate (EER) is commonly used because it is threshold-independent and directly comparable

across systems:

$$\text{EER} = P_{\text{FA}}(\tau^*) = P_{\text{FR}}(\tau^*) \quad (4.3)$$

where P_{FA} is the false-acceptance rate, P_{FR} is the false-rejection rate, and τ^* is the threshold at which the two rates are equal.

Prior work shows that spoofing artefacts are not confined to a single acoustic domain. Sub-band analyses showed that synthesis traces may appear in specific frequency regions [13, 14], while later studies showed that artefacts may be spectral, temporal, or both [7]. This finding is important because fixed representations such as MFCC, LFCC, CQT, and Mel-spectrograms impose a designer-chosen view of speech. Muller et al. [10] demonstrated that detectors often degrade sharply under cross-dataset conditions, and that Mel-spectrogram smoothing can remove high-frequency information useful for synthetic speech detection. A detector for Nepali should therefore avoid assuming that English spoofing artefacts and Nepali spoofing artefacts occupy the same acoustic regions.

4.1.2 Self-Supervised Feature Extraction with XLS-R

Self-supervised learning (SSL) addresses the limitation of handcrafted features by learning representations directly from raw audio before task-specific labels are introduced. XLS-R extends wav2vec 2.0 to a multilingual setting and was pre-trained on approximately 436,000 hours of speech from 128 languages [5]. This language coverage is the principal reason it is appropriate for a Nepali detector: the XLS-R language table includes Nepali with 58 hours of pre-training audio [5]. Consequently, adaptation to Nepali is not a fully from-scratch problem.

XLS-R follows the wav2vec 2.0 architecture. A convolutional feature encoder first converts a raw waveform into latent representations, and a Transformer context network then produces contextualised speech representations across time [5, 6]. During pre-training, masked time steps must be matched to their true quantised latent target against distractors, using a contrastive objective. A diversity term encourages the codebook to be used broadly rather than collapsing to a small set of entries. XLS-R also uses temperature-smoothed sampling to balance high-resource and low-resource languages during pre-training [5].

The project uses XLS-R because it is the model size used in the strongest published

wav2vec 2.0 and AASIST anti-spoofing reference system [9]. Table 1 summarises the three XLS-R model sizes reported by Babu et al. [5].

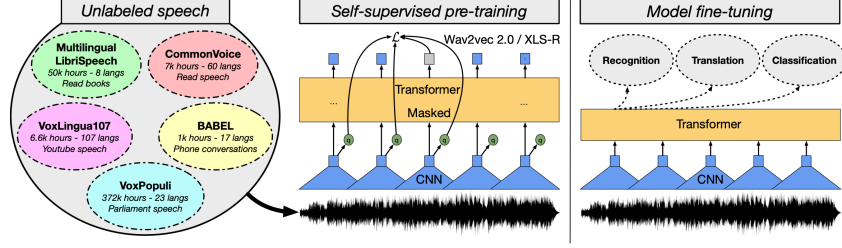


Figure 4.1: XLS-R self-supervised speech representation architecture used as the multilingual front-end.

Table 4.1: XLS-R model configurations reported in the literature

Model	Blocks	Hidden	FFN dim	Parameters
XLS-R 0.3B	24	1024	4096	317M
XLS-R 1B	48	1024	4096	965M
XLS-R 2B	48	1920	7680	2162M

The value of XLS-R for this project is threefold. First, its pre-training includes Nepali, unlike English-only wav2vec 2.0 models. Second, its learned representation is data-driven rather than locked to a fixed spectral transform. Third, Tak et al. [9] showed that an XLS-R front-end combined with an AASIST-style back-end can substantially improve audio spoofing detection, reducing ASVspoof 2021 Logical Access pooled EER from an AASIST-style baseline of 11.47% to 0.82% when combined with self-attentive aggregation and RawBoost augmentation.

4.1.3 Graph-Based Spectro-Temporal Modelling with AASIST

Graph attention networks (GATs) introduced masked self-attention over graph-structured data, allowing a node to learn how strongly it should attend to each neighbour rather than using a fixed aggregation rule [15]. For anti-spoofing, this property is valuable because spoofing artefacts may be concentrated in different spectral or temporal regions depending on the attack algorithm.

AASIST adapts this idea to audio anti-spoofing through integrated spectro-temporal graph attention [7]. Its original front-end is based on RawNet2 [8]. The resulting feature map is used to construct a spectral graph and a temporal graph. The Heteroge-

neous Stacking Graph Attention Layer (HS-GAL) then learns attention across spectral-spectral, temporal-temporal, and spectral-temporal edge types. A Max Graph Operation (MGO) uses parallel branches and element-wise maximum fusion so that different artefact-bearing patterns can compete. Finally, a readout stage combines node-wise maximum, node-wise average, and a stack node before classification [7].

AASIST achieved a best-seed EER of 0.83% and a minimum tandem Detection Cost Function (min t-DCF) of 0.0275 on ASVspoof 2019 Logical Access [7]. As shown in Table 2, it outperformed several strong raw-waveform and spectral baselines while using fewer parameters than many alternatives.

Table 4.2: Selected ASVspoof 2019 Logical Access results

System	Parameters	min t-DCF	EER (%)
AASIST [7]	297K	0.0275	0.83
AASIST-L [7]	85K	0.0309	0.99
RawGAT-ST [7]	437K	0.0335	1.06
SENet [?]	1,100K	0.0368	1.14
Res-TSSDNet [?]	350K	0.0481	1.64

In the proposed project, AASIST is used as the trainable back-end that receives contextualised XLS-R representations. This coupling is supported by Tak et al. [9], who validated wav2vec 2.0/XLS-R features with an AASIST-style back-end. The novelty in the present proposal is not the invention of a new graph-attention detector, but its careful adaptation to Nepali using parameter-efficient fine-tuning and a Nepali real/fake dataset.

4.1.4 Parameter-Efficient Fine-Tuning Mechanics

Full fine-tuning of XLS-R 0.3B would require updating approximately 317 million parameters, which is computationally expensive and risky for a small Nepali dataset. Two problems are especially relevant. First, catastrophic forgetting may overwrite the multi-lingual acoustic knowledge learned during pre-training. Second, overfitting may occur because the downstream Nepali dataset is far smaller than the 436,000 hours used for XLS-R pre-training. Parameter-efficient fine-tuning (PEFT) addresses these issues by freezing most of the backbone and training only a small number of injected parameters [21].

The PEFT literature includes additive methods such as adapters and prompt tuning, partial fine-tuning methods such as BitFit, reparameterised methods such as LoRA, and

hybrid methods [21, 22]. LoRA is particularly suitable here because it reparameterises the update to a frozen weight matrix as a low-rank product [20]. For a frozen attention projection $W_0 \in \mathbb{R}^{d \times k}$, LoRA learns:

$$\Delta W = W_{\text{down}} W_{\text{up}} \quad (4.4)$$

where $W_{\text{down}} \in \mathbb{R}^{d \times r}$, $W_{\text{up}} \in \mathbb{R}^{r \times k}$, and $r \ll \{d, k\}$. The adapted forward pass is:

$$h = W_0 x + s \cdot W_{\text{down}} W_{\text{up}} x \quad (4.5)$$

where s is a scaling factor. Only W_{down} and W_{up} are trained. At inference, the learned low-rank update can be merged into the original weight matrix, avoiding additional latency [20, 21].

This proposal combines LoRA with a wavelet-prompt adapter. The wavelet path is motivated by the fact that audio spoofing artefacts can be sub-band-localised and time-varying [10, 13, 14]. A wavelet decomposition provides multi-scale time-frequency localisation, while the learnable prompt gives the frozen XLS-R front-end an explicit signal about where Nepali-specific artefacts may occur. LoRA and wavelet prompting belong to different PEFT families, so they are complementary: LoRA modifies attention projections in a low-rank manner, while the wavelet prompt adds a trainable input-side cue.

4.2 Review of Related Works

4.2.1 Benchmark Datasets

Progress in audio deepfake detection has been driven by benchmark datasets. ASVspoof 2015, 2017, 2019, and 2021 established standard evaluation settings for spoofing countermeasures [1–4]. The 2019 Logical Access track remains particularly important because it contains both training and unseen evaluation attacks generated by multiple TTS and VC systems [3, 18]. English resources such as Fake-or-Real, WaveFake, and In-the-Wild have further expanded evaluation beyond challenge conditions [10].

The In-the-Wild dataset is especially significant because it revealed how poorly benchmark-trained detectors can generalise to real public audio [10]. Community-

maintained repositories also list many public audio deepfake detection resources, but the available tables do not indicate a public Nepali real-and-fake detection corpus [27, 28].

4.2.2 Detection Methods and Performance Milestones

Early systems relied on handcrafted acoustic features and conventional classifiers. Later work moved toward raw-waveform networks such as RawNet2 [8], graph-attention systems such as AASIST [7], and self-supervised front-ends such as wav2vec 2.0/XLSR [9]. Table 3 summarises the component-wise improvement reported by Tak et al. [9] on ASVspoof 2021 Logical Access.

Table 4.3: Effect of SSL front-end, aggregation, and augmentation in Tak et al.

Configuration	SSL front-end	Self-attentive aggregation	Pooled EER (%)
AASIST-style baseline	No	No	11.47
XLS-pR 0.3B frontend	Yes	No	6.15
XLS-R 0.3B + aggregation	Yes	Yes	4.48
XLS-R 0.3B + aggregation + RawBoost	Yes	Yes	0.82

These results justify using XLS-R plus AASIST as the reference architecture. However, they do not prove that the same detector will work for Nepali, because most benchmark evidence is English-centred.

4.2.3 Generalisation Collapse

The most persistent weakness in audio deepfake detection is generalisation. Muller et al. [10] re-evaluated multiple architectures under cross-dataset conditions and found large degradation when detectors trained on benchmark data were tested on in-the-wild audio. This shows that high in-distribution accuracy is not enough. The detector must be tested against audio whose language, platform, recording conditions, and synthesis process differ from the training set.

This point is directly relevant to Nepali. If a detector has learned English-specific spoofing traces, phonotactics, or recording conditions, there is no theoretical guarantee that the same score distribution will remain reliable for Nepali. The proposed project therefore treats English-to-Nepali transfer as an empirical question rather than an assumption.

4.2.4 Multilingual and Low-Resource Detection

Recent multilingual datasets have started to address the English and Chinese bias in the field. MLAAD, IndicFake, IndicSynth, SEA-Spoof, and BanglaFake broaden coverage to multilingual and South Asian settings, including languages such as Hindi, Bangla,

Marathi, Tamil, Urdu, and Bengali [?, ?, ?, ?, 12]. Nevertheless, the project draft and available dataset summaries indicate that Nepali is still absent from these resources.

Table 4.4: Language coverage gap in relevant audio deepfake resources

Dataset or resource	Language coverage	Nepali included
ASVspoof 2019/2021	Primarily English	No
In-the-Wild	Public English celebrity/politician audio	No
MLAAD	38+ languages	No reported Nepali
IndicFake	English plus 17 Indian languages	No reported Nepali
IndicSynth	12 low-resource Indian languages	No reported Nepali
SEA-Spoof	Six South-East Asian languages	No
BanglaFake	Bengali	No
Proposed project	Nepali real and fake speech	Yes

4.2.5 Nepali-Specific and Nepal-Originated Work

The Nepali literature shows a generation-detection imbalance. On the generation side, Nepali voice cloning and Nepali TTS work has emerged, including transfer-learning-based voice cloning, few-shot multi-speaker cloning, and Tacotron-style Nepali synthesis [23–25]. On the detection side, the closest Nepal-originated work is the CNN-BiLSTM detector with attention by Chapagain et al. [26]. That work is important locally, but its dataset is English: SceneFake and Fake-or-Real. It therefore does not solve Nepali-language detection.

4.3 Research Gap

The literature establishes four facts. First, synthetic speech generation is now mature enough to create convincing low-resource-language voice deepfakes. Second, XLS-R plus AASIST is a strong and empirically validated architecture for anti-spoofing under benchmark conditions [7, 9]. Third, benchmark performance does not guarantee real-world or cross-lingual robustness [10]. Fourth, realistic in-domain fine-tuning is one of the most practical remedies for generalisation collapse.

Three specific gaps follow from these facts. The first gap is language coverage bias. Existing datasets are dominated by English and a limited number of other languages, while Nepali remains unrepresented in the detection resources reviewed. The second gap is the lack of evidence for English-to-Nepali detector transfer. Since cross-lingual degradation is common, an English-trained model cannot be assumed to protect Nepali speakers. The third gap is the absence of a public Nepali bona fide and spoofed audio

detection resource. Existing Nepali work supports the generation side of the problem, while detection remains underdeveloped.

CHAPTER 5

PROPOSED METHODOLOGY

5.1 Project Approach

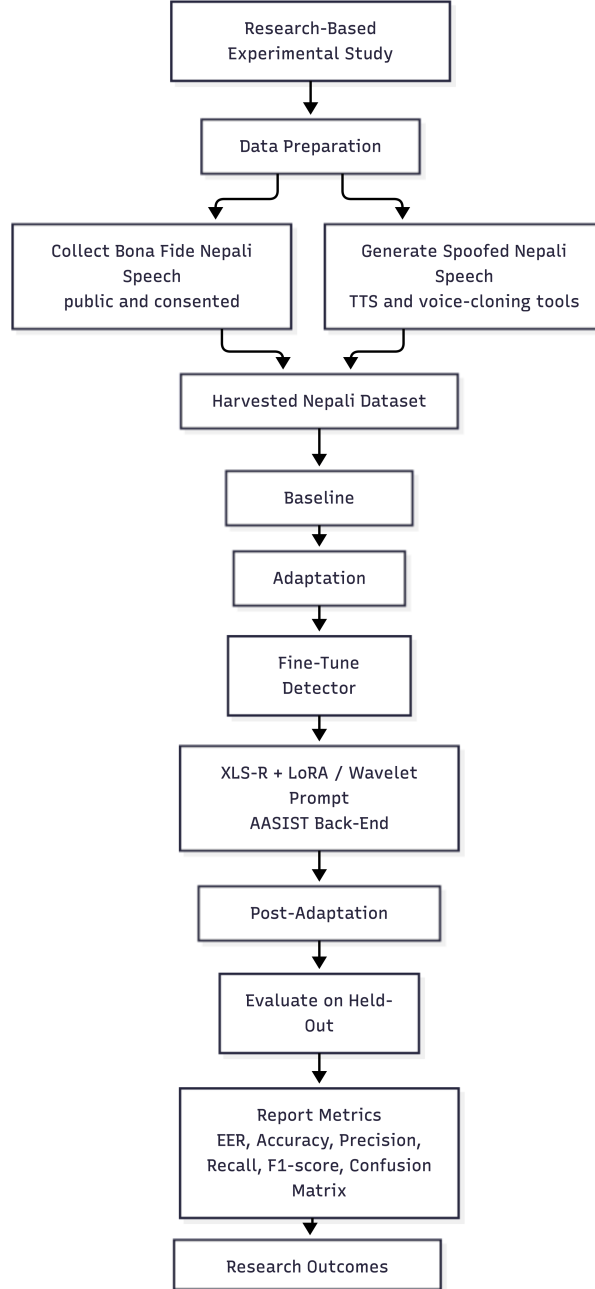


Figure 5.1: Project approach and experimental pipeline.

A systematic project design is proposed to establish a robust framework for audio deep-fake detection in Nepali linguistic domain. The approach aims at shifting the massive architectures toward highly targeted, language-specific acoustic modelling using param-

eter efficient optimization. The project is structured into three distinct phases: Data collection and verification, parameter-efficient fine tuning and robust evaluation.

To maximize structural flexibility and minimize training, volatility, a decoupled processing strategy is employed. The system cleanly separated the high-dimensional feature representation task from the downstream structural classification task. By isolating the front end acoustic encoder from the backend graph classifier distinct optimization constraints can be applied to each module independently, preventing the optimization of classification boundaries from corrupting the generalized speech patterns preserved within the base transformer blocks.

5.2 System Architecture & Design Framework

The proposed system architecture integrates a multilingual self-supervised front-end, parameter-efficient adaptation, and a graph-attention back-end [5, 7, 9, 21]. An overview of the framework is presented in Figure 5.2.

The framework operates through the following stages:

- **Input:** A Nepali audio clip.
- **Preprocessing:** The raw audio is converted to mono, resampled to 16 kHz, and subjected to silence trimming and amplitude normalisation. Clips are segmented or padded to a fixed duration to ensure uniform input shape [7, 9].
- **Front-End:** XLS-R 0.3B is utilised as the multilingual self-supervised feature extractor. The backbone is primarily frozen to retain its pre-trained acoustic knowledge [5].
- **Adaptation Layer:** Low-Rank Adaptation (LoRA) adapters are inserted into the attention projection layers of the XLS-R encoder. Concurrently, a wavelet prompt branch is applied to provide time-frequency and sub-band cues to the frozen front-end [?, 13, 20, 21].
- **Back-End:** The AASIST graph-attention back-end receives the adapted XLS-R-derived features. It performs spectral and temporal graph modelling to capture spoofing artefacts [7].

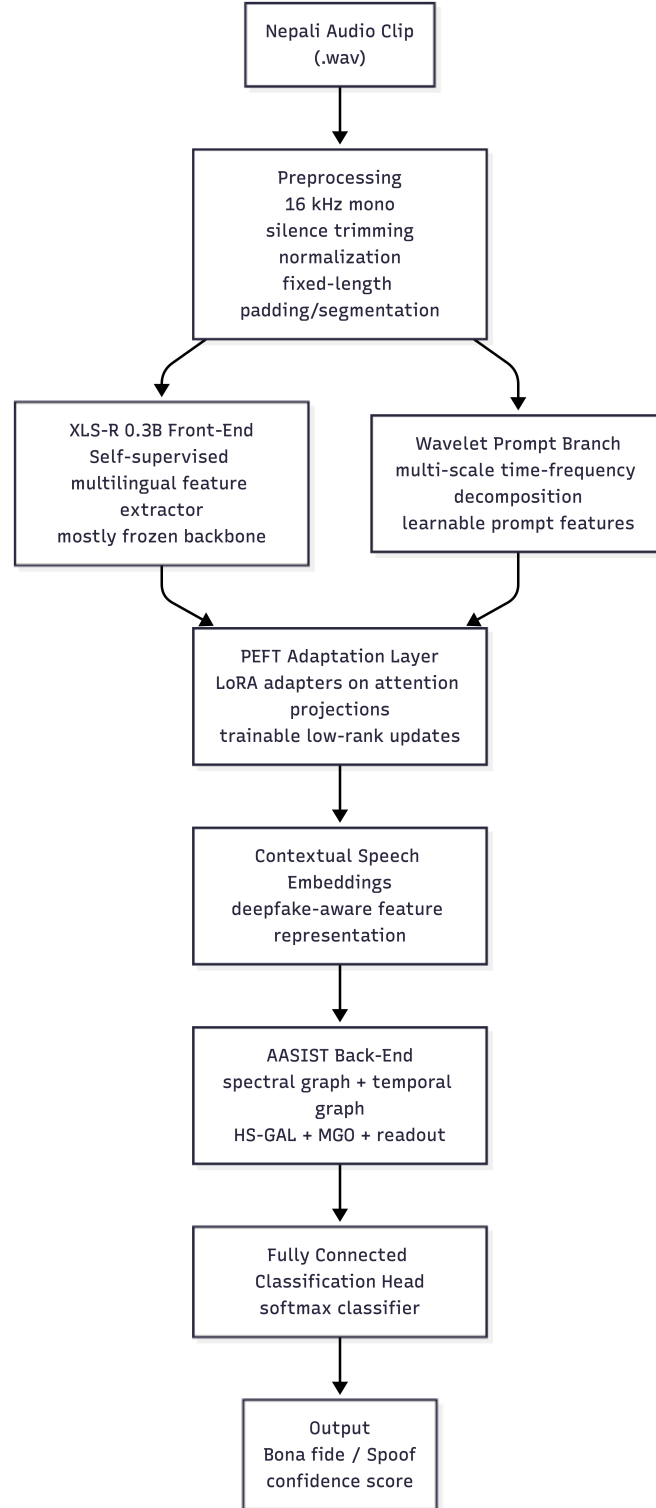


Figure 5.2: Proposed XLS-R 0.3B, PEFT, AASIST, and classification framework for Nepali audio deepfake detection.

- **Classification:** A fully connected classification head processes the AASIST graph readouts to produce a final output. The system yields a bona fide/spoof probability and a corresponding confidence score.

5.3 Algorithms and Models to be Used

This subsection details the specific algorithms and model components employed in the proposed framework.

5.3.1 XLS-R

XLS-R 0.3B is utilised to extract multilingual contextual speech representations from Nepali audio. Because XLS-R includes Nepali in its pre-training corpus, it is more suitable for this task than English-only wav2vec 2.0 models [5]. In the proposed framework, XLS-R 0.3B serves as the frozen or PEFT-adapted front-end feature extractor [9].

5.3.2 LoRA

Low-Rank Adaptation (LoRA) is applied to adapt the XLS-R front-end without requiring full fine-tuning. LoRA learns a low-rank update to selected attention projection matrices, specifically targeting the query and value projections [20]. This approach drastically reduces the number of trainable parameters and helps prevent catastrophic forgetting of the multilingual acoustic knowledge acquired during pre-training [21].

5.3.3 Wavelet Prompt

A wavelet prompt mechanism is incorporated to provide multi-scale time-frequency information for spoof artefact detection. Because audio spoofing artefacts may manifest in specific sub-bands and temporal regions, wavelet decomposition helps localise these anomalies more effectively than a single fixed spectral view [?, 13, 14]. The learnable prompt provides the frozen XLS-R encoder with explicit cues regarding where Nepali-specific artefacts may occur.

5.3.4 AASIST Back-End

The Audio Anti-Spoofing using Integrated Spectro-Temporal graph attention networks (AASIST) serves as the trainable back-end. AASIST models spectral and temporal relationships through graph attention, receiving the XLS-R-derived features [7]. This component supports artefact modelling across both frequency and time dimensions, enabling robust detection of synthesis traces.

5.3.5 Classification Head

A fully connected classification head is used to produce the final real/fake prediction. The output consists of the probabilities $P(\text{bona fide})$ and $P(\text{spoof})$, the predicted label, and an associated confidence score.

Table 5.1: Model components and their roles in the proposed framework

Component	Role	Trainable	Justification
XLS-R 0.3B	Feature extraction	Frozen/PEFT	Multilingual SSL, includes Nepali
LoRA	Efficient adaptation	Yes	Low trainable parameter count
Wavelet prompt	Artefact localisation	Yes	Time-frequency/sub-band cues
AASIST	Graph back-end	Yes	Spectro-temporal spoof modelling
FC head	Classification	Yes	Real/fake output

5.4 Data Acquisition

The dataset acquisition plan is critical to addressing the literature gap, as existing audio deepfake detection resources remain concentrated around a limited set of benchmark languages and do not provide sufficient Nepali language coverage [12, 16, 17].

5.4.1 Bona Fide Nepali Speech

Bona fide Nepali speech will be collected from public sources, including public Nepali interviews, speeches, news clips, podcasts, and educational recordings. Voluntary recorded samples may also be utilised if permitted by the project scope. The collection will prioritise speaker labels, gender and speaker diversity, and sufficient audio clarity for segmentation. All public or consented data will be used in accordance with ethical use statements, since recent surveys emphasise dataset transparency and real-world representativeness as continuing challenges in audio deepfake detection [16, 17].

5.4.2 Spoofed Nepali Speech

Spoofed Nepali speech will be generated using Nepali TTS tools and voice cloning systems. Where available, multi-speaker synthetic samples will be produced. The generation process will use the same or comparable text prompts across different synthesis systems. Metadata detailing the generation method will be retained for each sample, since TTS, voice conversion, few-shot cloning, and zero-shot cloning represent the major families of speech deepfake generation [17, 19].

5.4.3 Preprocessing

All audio clips will undergo consistent preprocessing prior to model input. This includes resampling to 16 kHz, mono conversion, silence trimming, and peak or RMS normalisation. Fixed-length segmentation will be applied, and any corrupted or extremely noisy files will be removed from the dataset. Consistent preprocessing is required to reduce nuisance variation before baseline and adapted-model comparison [7, 9].

5.5 Evaluation Metrics

The proposed system will be assessed for efficiency and reliability through the use of a suite of performance metrics. These metrics are widely adopted in the ASVspoof Challenges and recent audio deepfake detection literature to assess both detection accuracy and robustness [3, 4, 16]. The proposed evaluation metrics are:

5.5.1 Equal Error Rate (EER)

Equal Error Rate (EER) is the principle evaluation metric in audio anti-spoofing research and has been extensively used in the ASVspoof challenges. EER represents the operating point at which the False Acceptance Rate (FAR) becomes equal to the False Rejection Rate (FRR).

The False Acceptance Rate is defined as:

$$\text{FAR} = \frac{FP}{FP + TN} \quad (5.1)$$

where

- FP = Number of spoofed samples incorrectly classified as genuine
- TN = Number of spoofed samples correctly classified

Similarly, the False Rejection Rate is defined as:

$$\text{FRR} = \frac{FN}{TP + FN} \quad (5.2)$$

where

- FN = Number of genuine samples incorrectly classified as spoofed

- TP = Number of genuine samples correctly classified

The equal error rate is obtained at the threshold where:

$$FAR = FRR \quad (5.3)$$

5.5.2 Classification Accuracy

Classification accuracy measures the proportion of correctly classified samples among the total number of evaluated samples. It is computed as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (5.4)$$

where

- TP = True Positives
- TN = True Negatives
- FP = False Positives
- FN = False Negatives

Accuracy provides a straightforward measure of overall classification performance.

5.5.3 F1-Score

The F1-score is a performance metric that provides a balanced evaluation between precision and recall. It is particularly useful in audio deepfake detection scenarios where the distribution of genuine and spoofed samples may not always be balanced. Unlike accuracy, which can be biased toward the majority class, the F1-score considers both false positive and false negative errors, providing a more reliable measurement of detection performance.

The F1-score is calculated as:

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (5.5)$$

where Precision and Recall are defined as:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (5.6)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (5.7)$$

where:

- TP = Number of spoofed samples correctly classified as spoofed
- FP = Number of genuine samples incorrectly classified as spoofed
- FN = Number of spoofed samples incorrectly classified as genuine

5.5.4 Confusion Matrix

A confusion matrix is used to analyze the detailed classification behavior of the proposed model. It provides information about the number of correctly and incorrectly classified samples and helps identify specific failure cases of the detection system.

For binary audio deepfake detection, the confusion matrix is represented as a tabular form showing the relationship between actual and predicted classes.

Table 5.2: Confusion Matrix for Binary Audio Deepfake Detection

	Predicted Class	
	Genuine	Spoof
Actual Genuine	TP (True Positive)	FP (False Positive)
Actual Spoof	FN (False Negative)	TN (True Negative)

where:

- **TP (True Positive)**: Genuine samples correctly classified as genuine
- **FP (False Positive)**: Genuine samples incorrectly classified as spoof
- **FN (False Negative)**: Spoof samples incorrectly classified as genuine
- **TN (True Negative)**: Spoof samples correctly classified as spoof